

INDIA'S AGRICULTURAL CROP PRODUCTION DASHBOARD

Project Report

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Agricultural data covering 1997–2021

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1. Introduction

Agriculture is one of the most important parts of India's economy and food system. Its performance is affected by cultivated area, crop choice, productivity and regional conditions. Because these factors change across years and states, a dashboard is useful for seeing the bigger picture.

The purpose of this project is to study an agricultural performance dashboard and explain the patterns shown in it in simple terms. The report does not treat a chart as an answer by itself. Instead, the charts are read together so that changes in production, yield and cultivated area can be understood in context.

2. Objectives of the Project

- To understand the overall agricultural performance shown by the dashboard.
- To examine how agricultural yield changes over the period 1997–2021.
- To compare cultivated area across major crops and states.
- To observe differences in agricultural production between states.
- To study the seasonal distribution of production in the available dataset.
- To interpret the dashboard from farmer, investor and policy-analyst viewpoints.

3. Dashboard and Data Overview

The analysis is based on the values and visualisations displayed in the supplied Tableau dashboard screenshots. The dashboard contains data for crop years from 1997 to 2021 and uses measures such as production, cultivated area and average yield. The main visual forms used are bar charts, a line chart, a map, a donut chart and summary cards. The observations in this report are descriptive and are limited to what is visible in the dashboard.

The summary card in the dashboard shows an average yield of 8,487.41, cultivated area of 4,031.13 M, and production of 330,997.84 M. These figures are reproduced as displayed in the dashboard and should be read with the dashboard's own units and aggregation settings.

India Agricultural Performance Dashboard

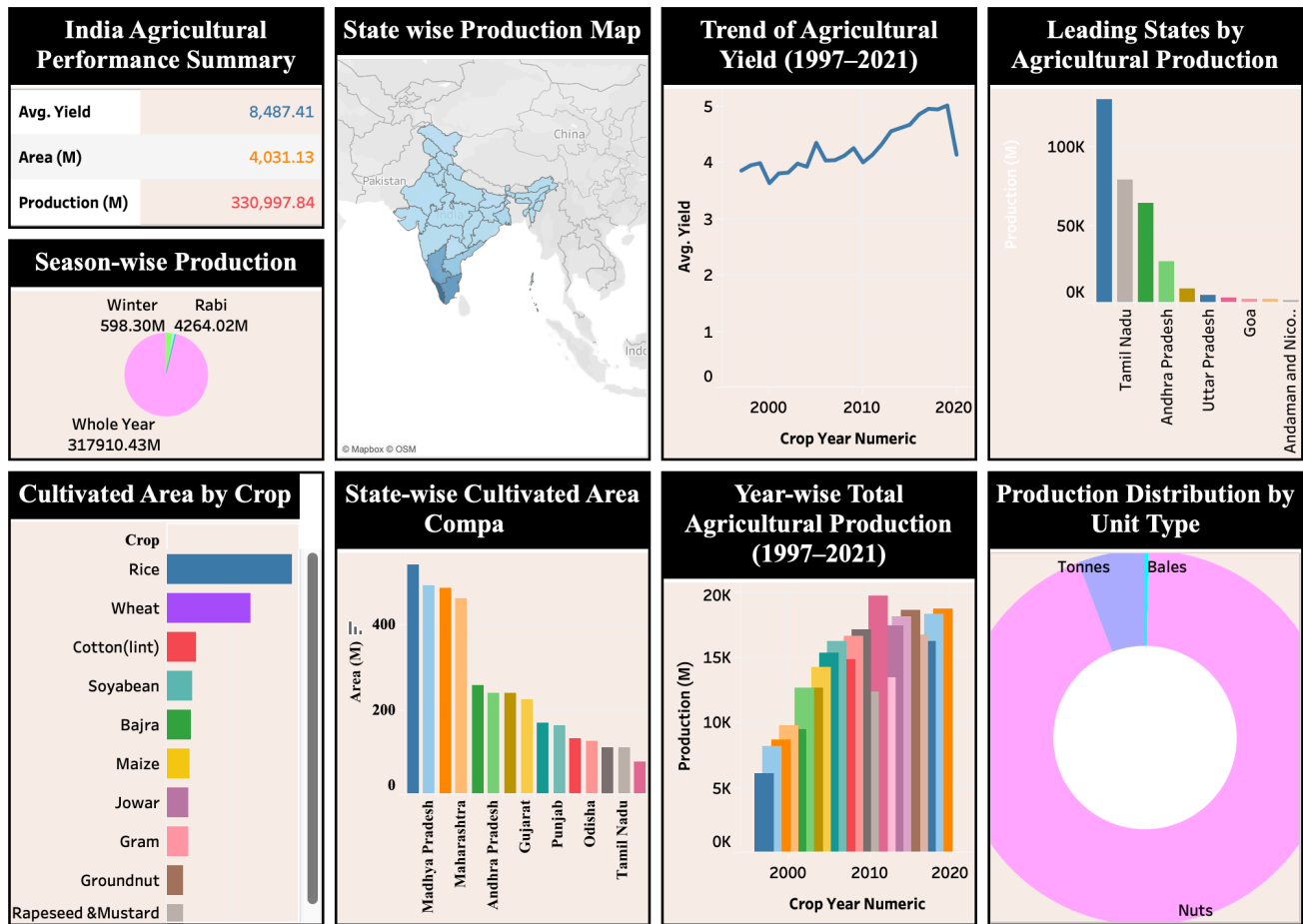


Figure 1: Overall India Agricultural Performance Dashboard.

4. Overall Agricultural Performance

The dashboard gives a quick picture of the dataset through three headline measures: average yield, cultivated area and production. The map and state-wise charts then add a regional dimension to these totals.

- The average-yield line shows a general upward movement over the long period, although there are several short-term falls.
- The yield reaches around 5 on the chart near 2019 before dropping sharply in the final plotted year, 2020.
- The production-by-year chart shows considerable variation, with several years reaching above 18K and a visible peak around 2011.
- The state production map uses shading to show differences between states, making the regional concentration of production easy to see.

4.1 Trend of Agricultural Yield

The yield line starts at roughly 3.8 in 1997. It falls to about 3.6 around 2000, then gradually moves upward. There is a noticeable rise around 2005, another dip around 2010, and a stronger upward movement after 2011. The line reaches about 5.0 in 2019 and then falls to a little above 4.0 in 2020. The important point is that the long-term direction is upward even though the yearly path is not smooth.

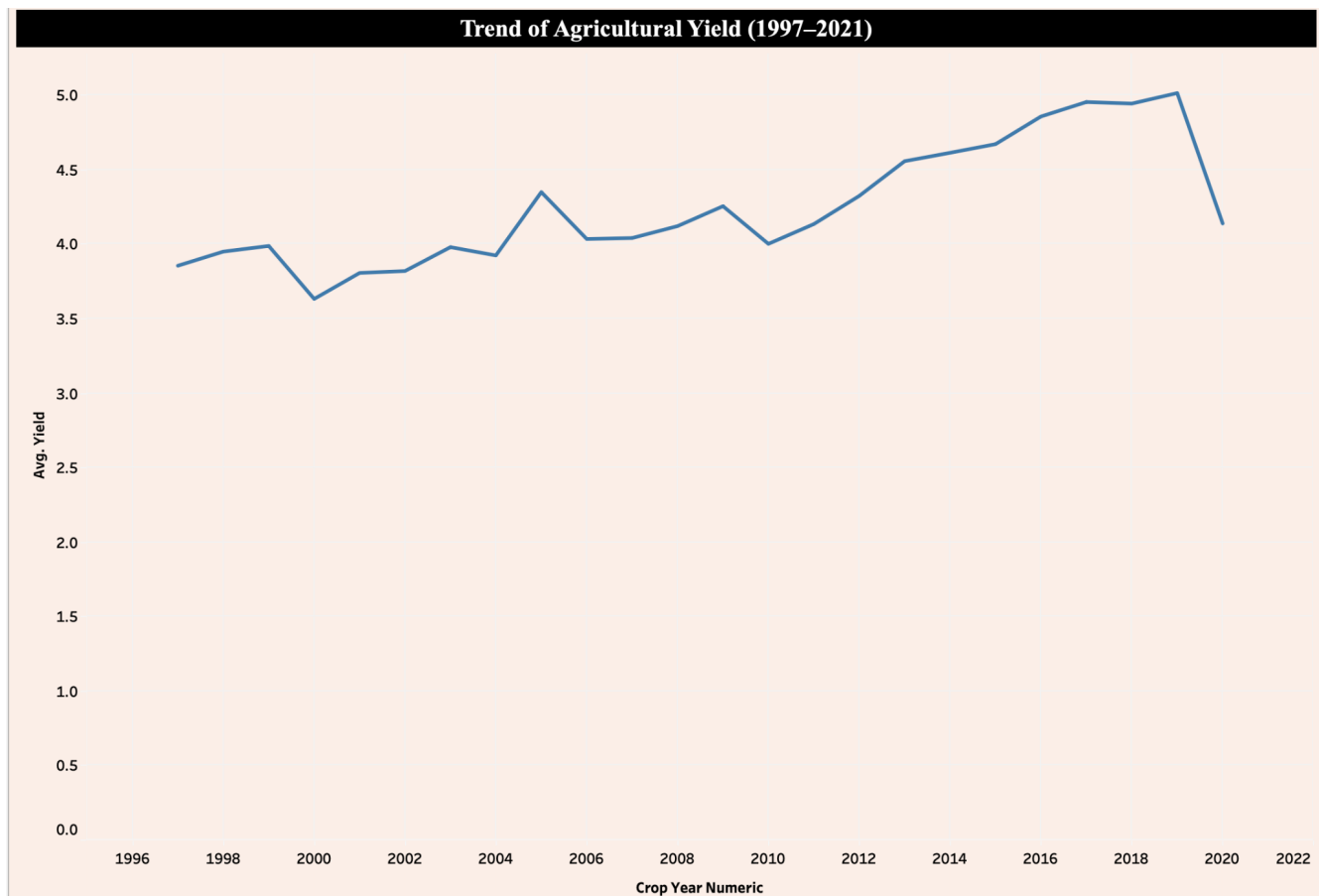


Figure 2: Trend of Agricultural Yield (1997–2021).

4.2 Year-wise Agricultural Production

The year-wise production chart shows that production does not rise in a straight line. It increases from the late 1990s, reaches high levels in the mid-2000s and again during the 2010s. The highest visible bar is around 2011, at just below 20K, while later years also remain comparatively high.

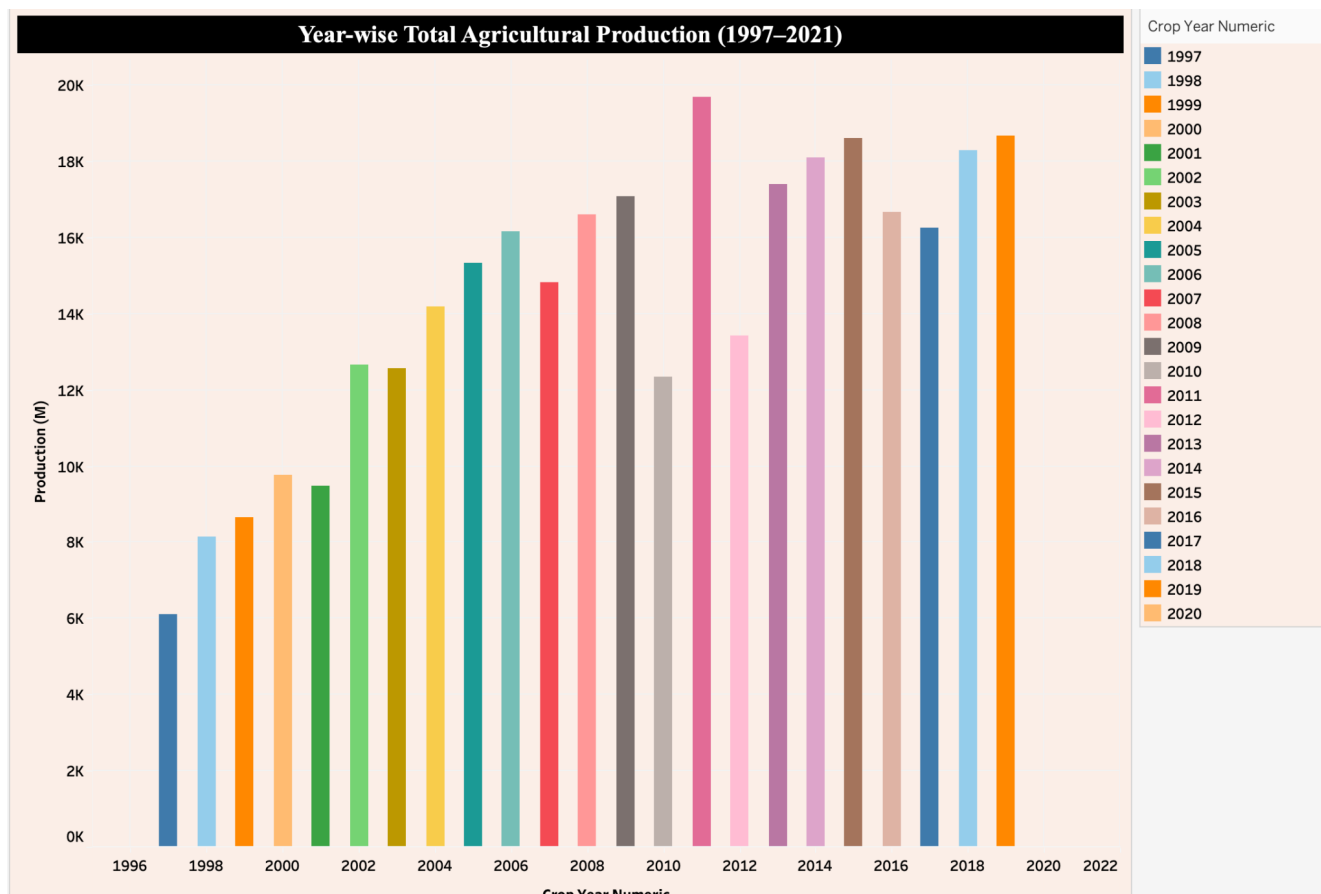


Figure 3: Year-wise Total Agricultural Production (1997–2021).

4.3 State-wise Production Map

The map gives a geographical view of production. Darker shading represents higher production according to the legend, while lighter shading represents lower production. The visual makes regional differences much easier to notice than a table of state names alone.

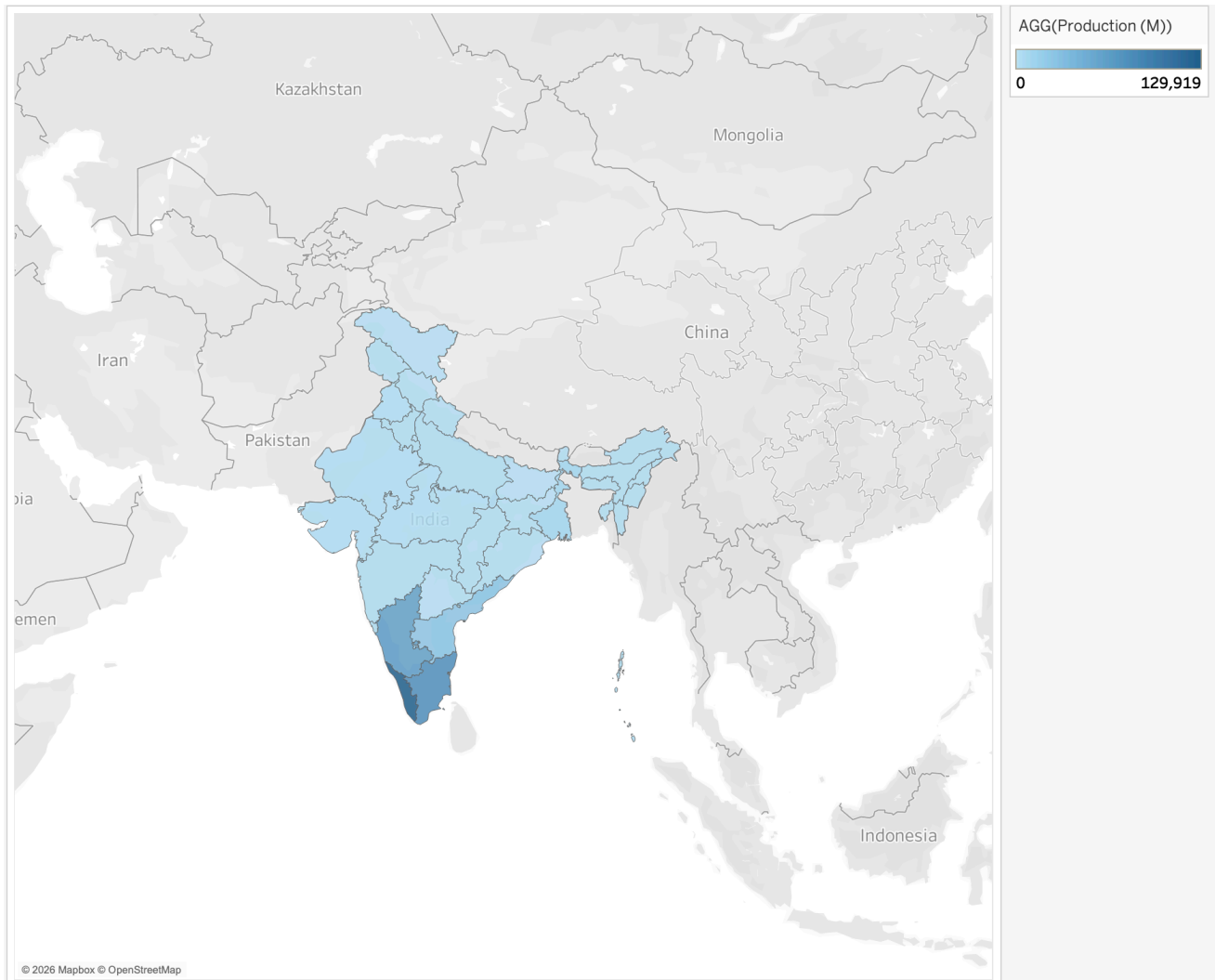


Figure 4: State-wise Production Map.

5. Farmer's View — Seasonal & Crop Insights

For a farmer, the most useful parts of the dashboard are the crop-wise cultivated area, the seasonal production chart and the yield trend. Together they give a basic picture of crop concentration and the way production changes over time.

5.1 Season-wise Production

The season chart is heavily dominated by the 'Whole Year' category in the displayed dataset. It is labelled 317910.43M, while Rabi is 4264.02M and Winter is 598.30M. This should not automatically be read as the physical amount produced in one agricultural season. It reflects how the dataset records the season field, and therefore the 'Whole Year' category needs to be understood as a separate category in the data.

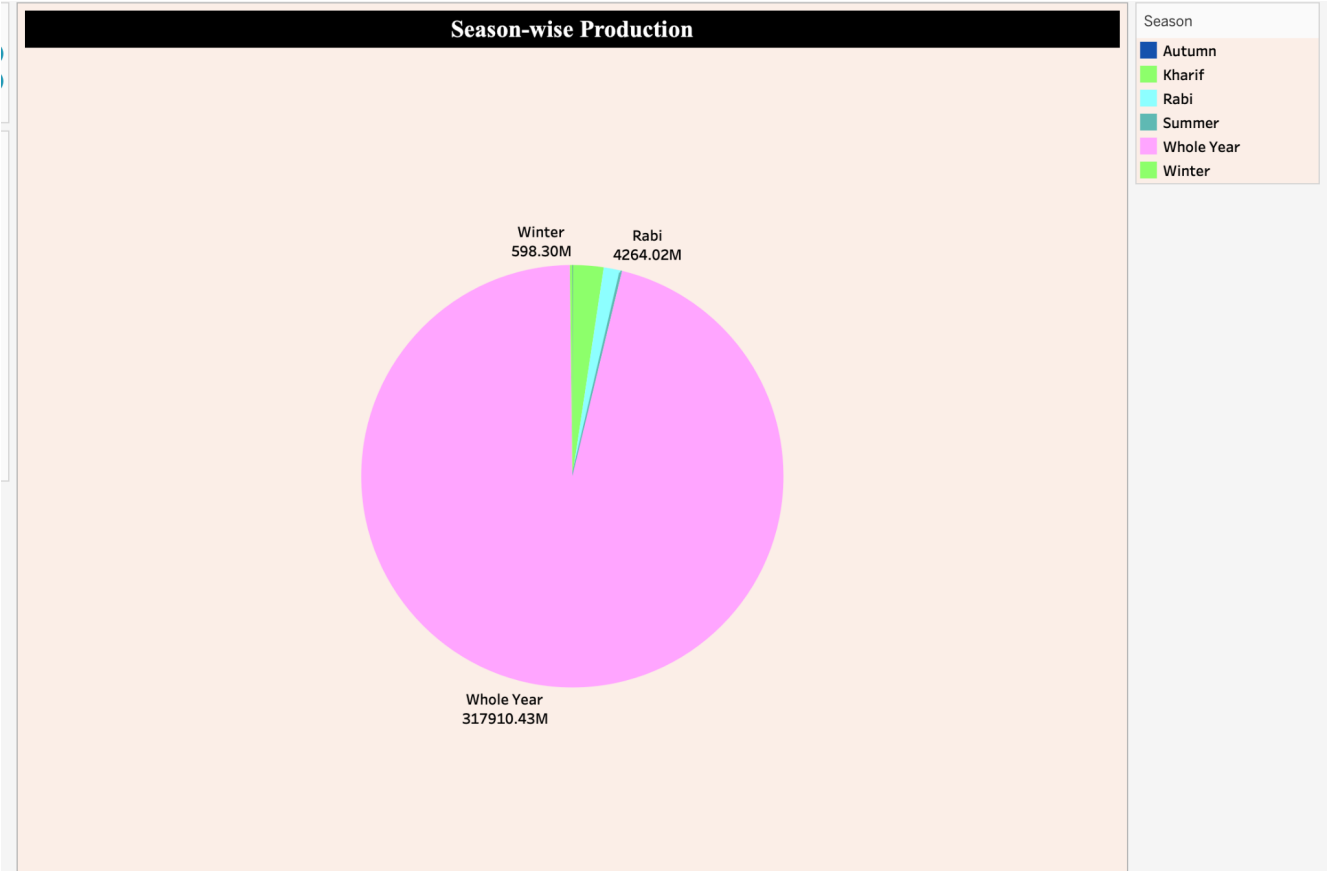


Figure 5: Season-wise Production.

5.2 Cultivated Area by Crop

Rice occupies the largest cultivated area in the displayed crop comparison, followed by wheat. Cotton, soybean and bajra form the next group, while maize, jowar, gram, groundnut and rapeseed & mustard appear with smaller areas. The large gap between rice and most other crops shows how concentrated the cultivated area is among a few crops.

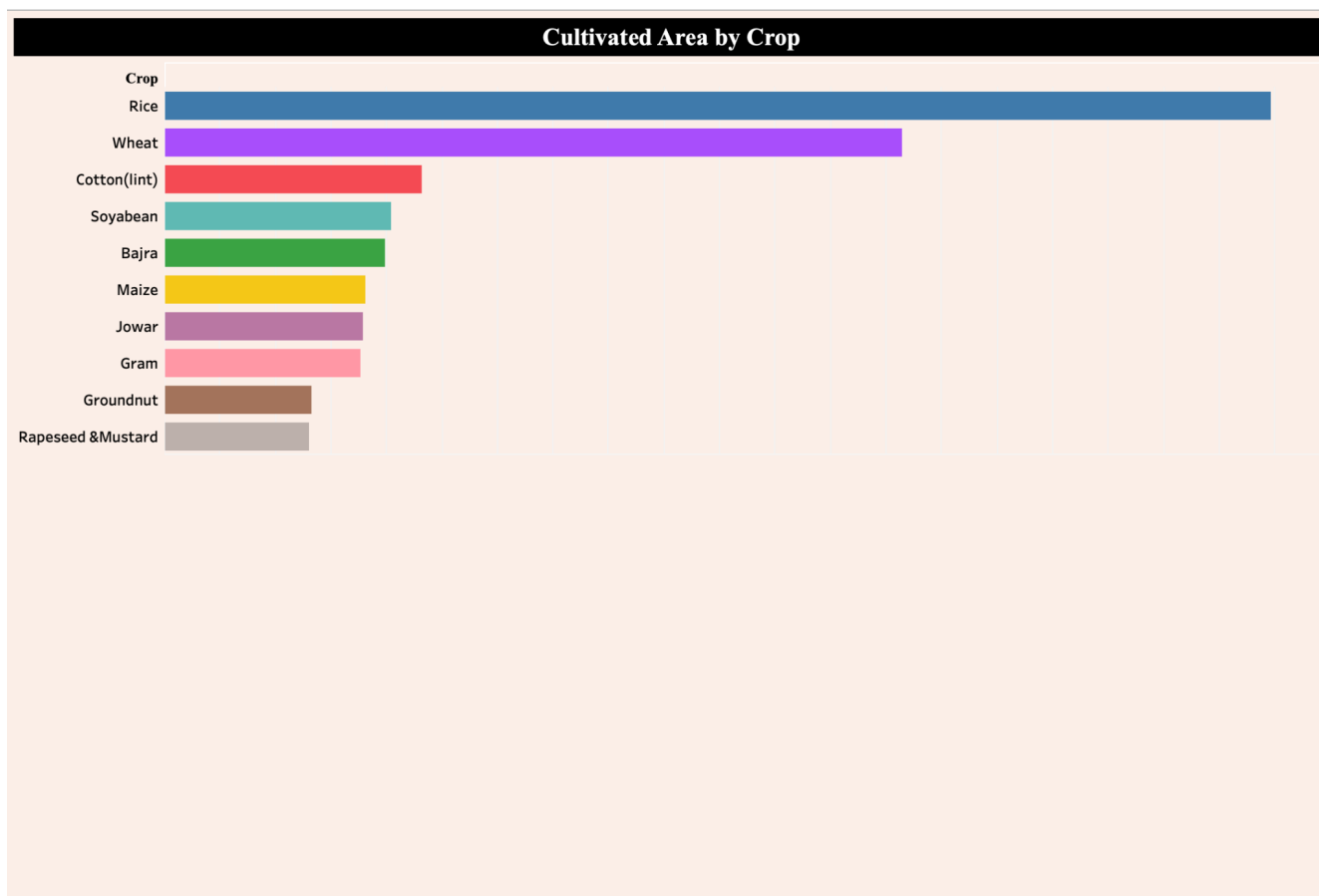


Figure 6: Cultivated Area by Crop.

5.3 What the Farmer View Suggests

- Crop concentration is important: rice and wheat occupy substantially more area than the other crops shown.
- Yield has improved over the long period, but the annual movement contains several dips.
- The dashboard can be used to compare crop patterns, but a real farm decision would also require local prices, rainfall, irrigation, soil and input costs.

6. Investor's View — Growth & Opportunity Insights

From an investor's point of view, the useful signals are the scale of production, its movement over time and the concentration of production among states. The dashboard does not provide profitability or share-price information, so the observations here are about the agricultural production data itself.

6.1 Production Growth Over Time

The year-wise chart shows an overall increase in production compared with the late 1990s. Production crosses the 18K level in a number of years, and the highest visible point is around 2011 at close to 20K. Later years remain well above the earliest values, indicating a larger production scale in the later part of the series.

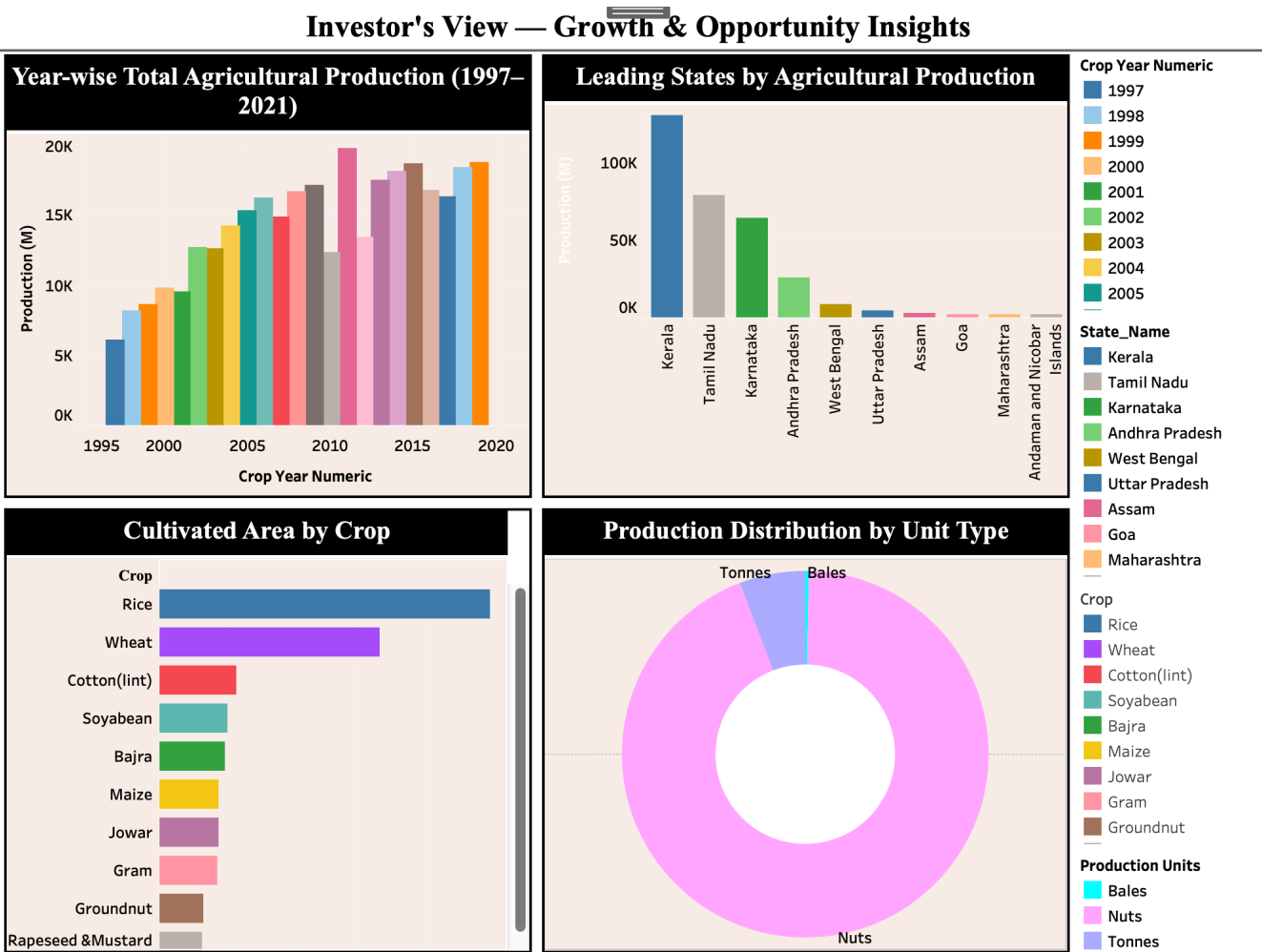


Figure 7: Investor View — Year-wise Total Agricultural Production.

6.2 Leading States by Agricultural Production

In the state comparison displayed in the dashboard, Kerala has the largest bar at approximately 129,919 M, followed by Tamil Nadu at roughly 79,000 M and Karnataka at about 64,000 M. Andhra Pradesh is the next major contributor at around 26,000 M. The remaining states shown are much smaller in comparison.

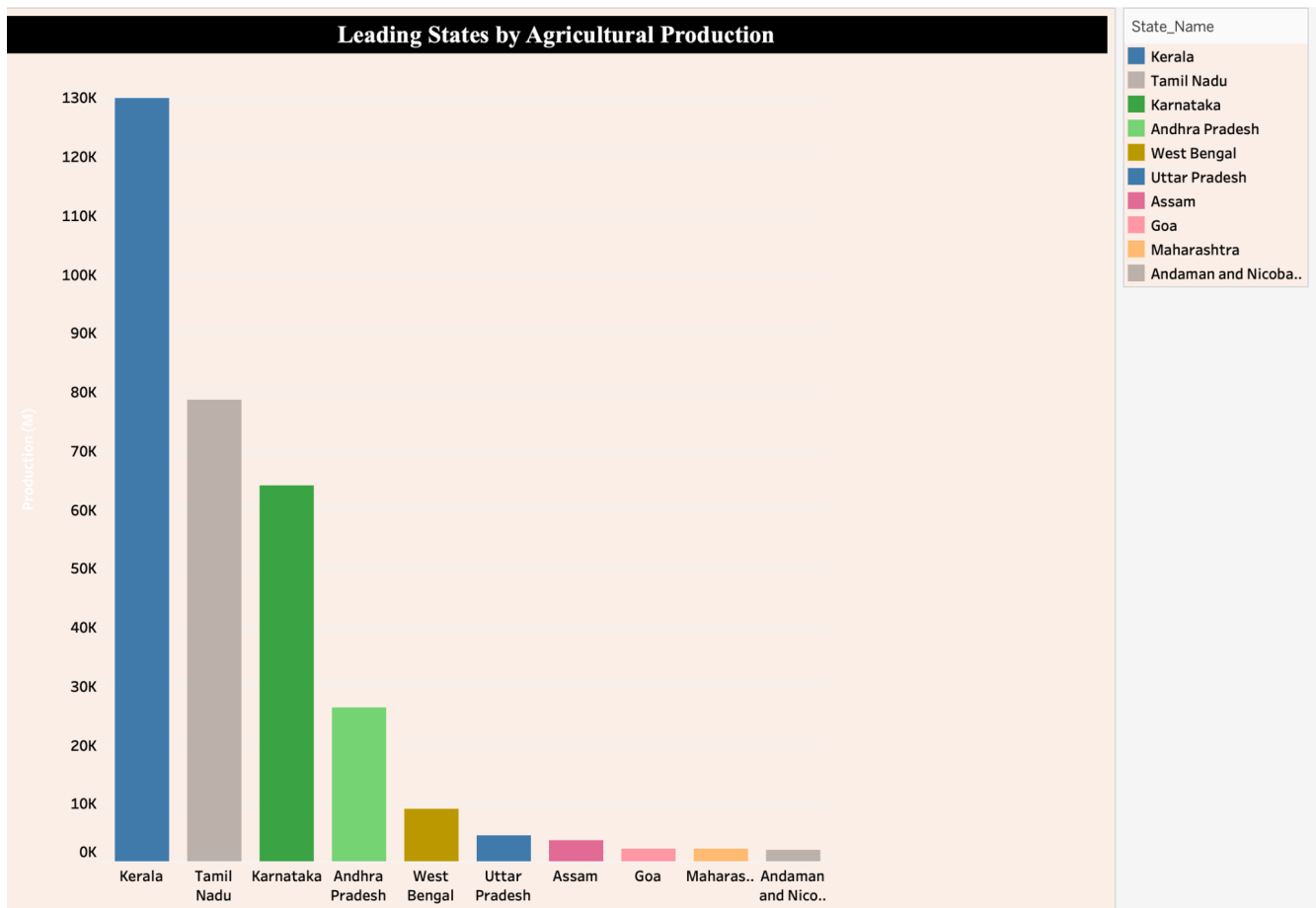


Figure 8: Leading States by Agricultural Production.

6.3 Production Distribution by Unit Type

The unit-type donut chart is dominated by the 'Nuts' category. Tonnes form a much smaller share and Bales are only a very small slice. This is a useful reminder that the dashboard combines records reported under different production units; the unit field therefore matters when interpreting totals.

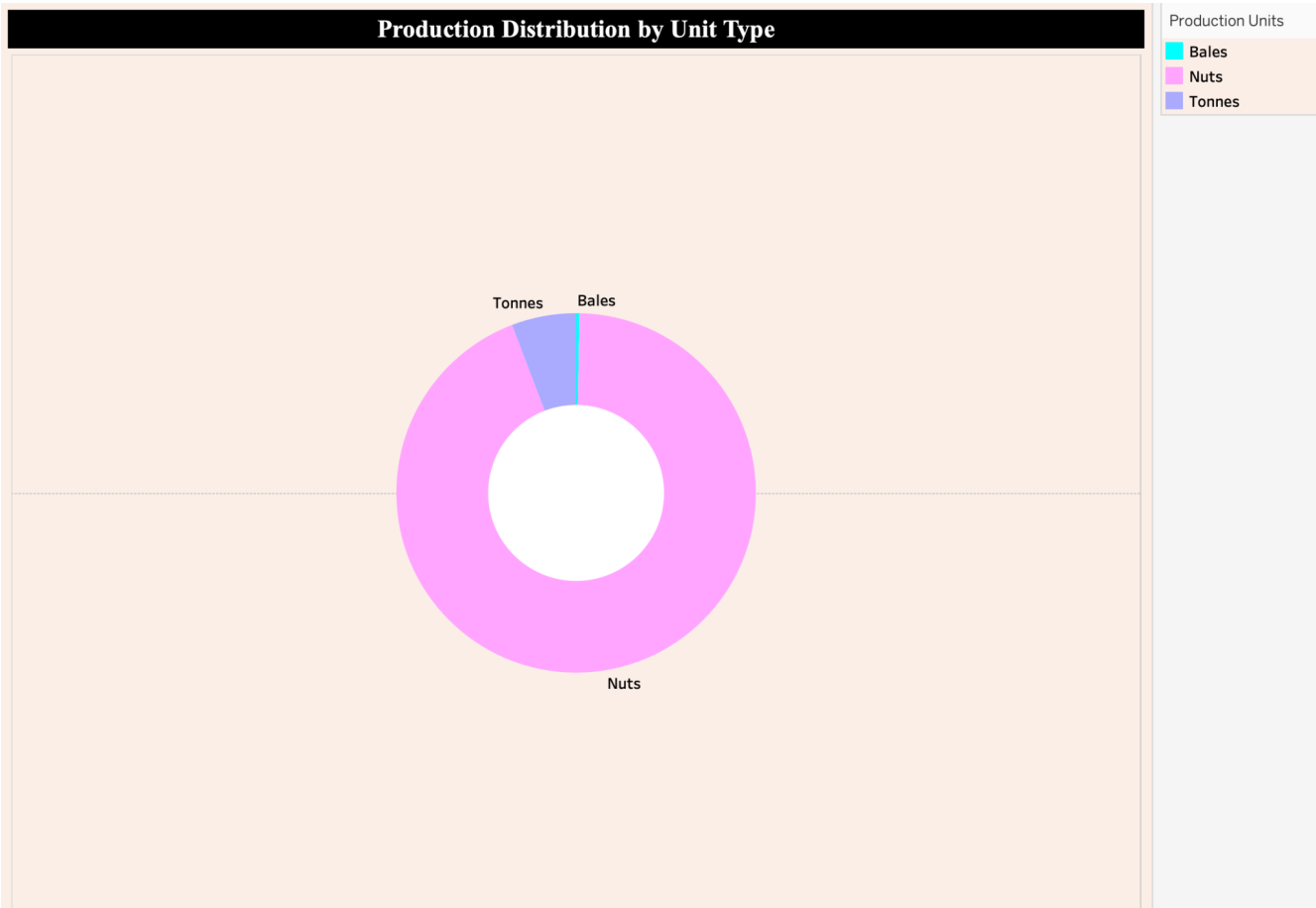


Figure 9: Production Distribution by Unit Type.

7. Policy Analyst's View — Regional Performance & Trends

The policy view brings regional differences together. It combines the production map, state-wise cultivated area, yield trend and leading-state production chart. This is useful for identifying where agricultural activity is more concentrated and where the differences between states are large.

7.1 State-wise Cultivated Area

The state cultivated-area chart shows Uttar Pradesh at about 540 M, followed by Madhya Pradesh near 490 M, Rajasthan near 485 M and Maharashtra around 460 M. Karnataka, Andhra Pradesh and West Bengal form a second group around the 230–260 M range. Assam is the smallest among the states displayed, at roughly 75 M.

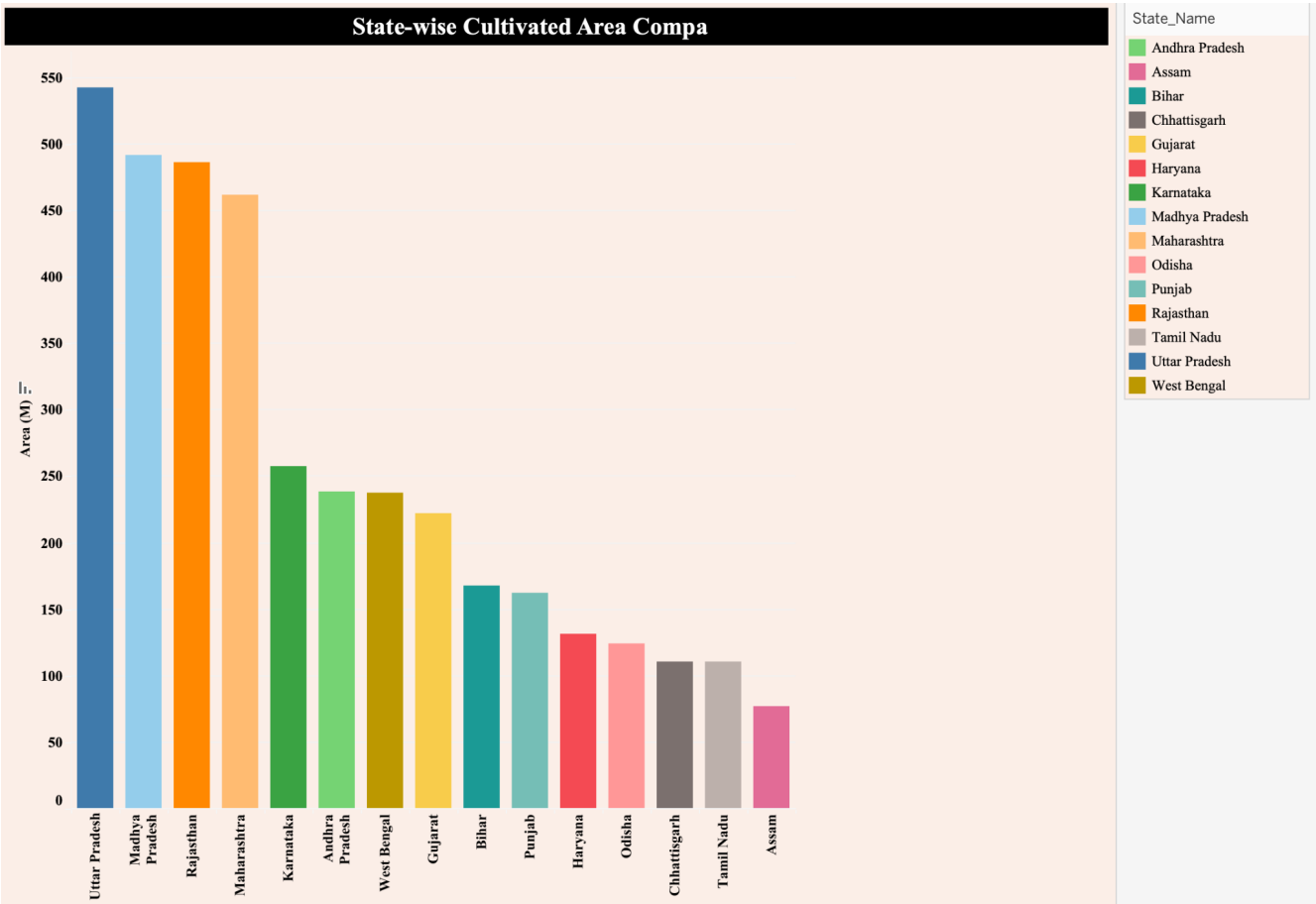


Figure 10: State-wise Cultivated Area Comparison.

7.2 Regional Production Pattern

The production comparison shows a strong difference between the largest and smaller contributors. Kerala's displayed production is around 130K M, while Tamil Nadu is near 80K M and Karnataka around 64K M. The gap between these values and the smaller states is quite large. The map provides the same information spatially rather than numerically.

Policy Analyst's View — Regional Performance & Trends

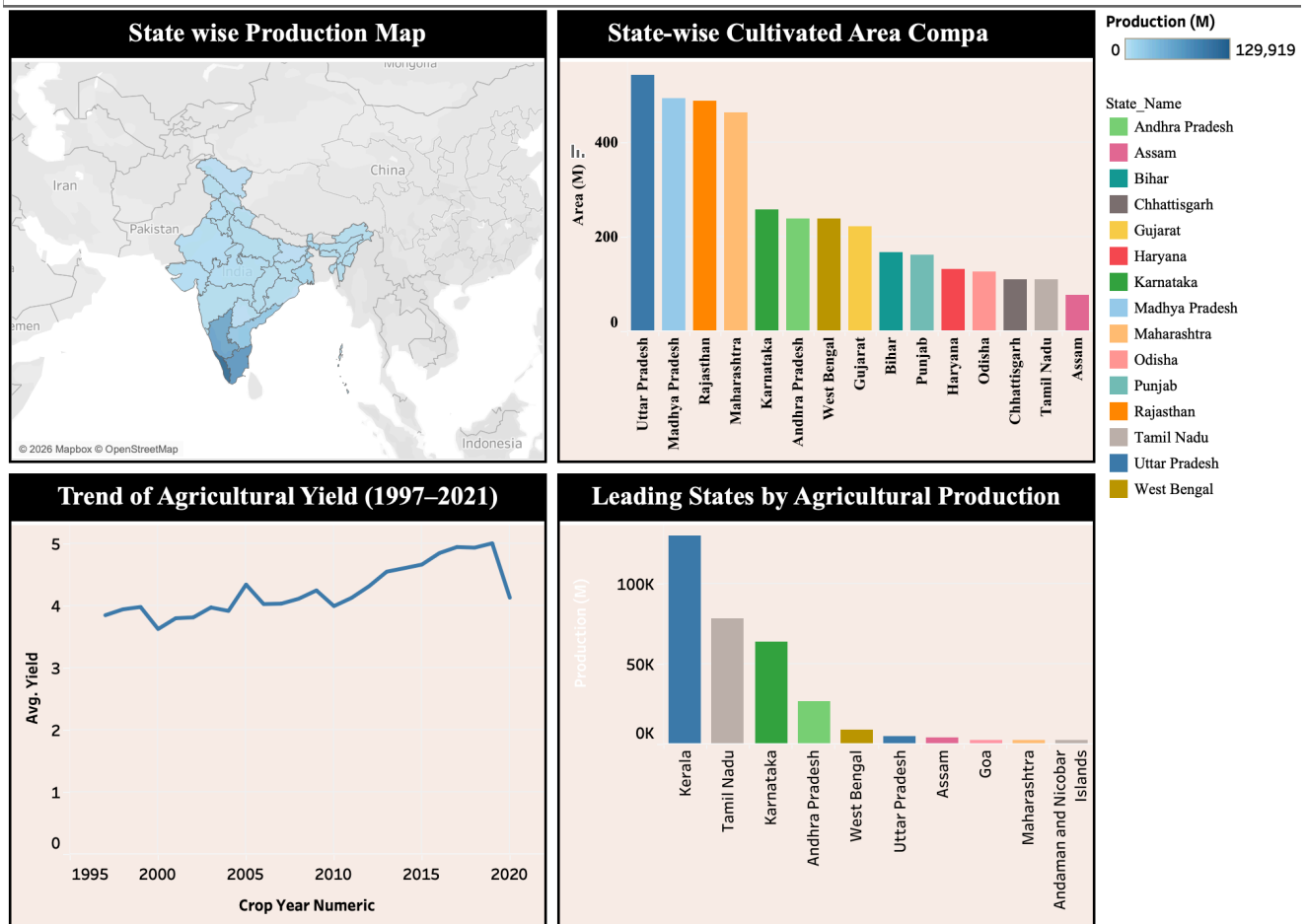


Figure 11: Policy Analyst's View — Regional Performance & Trends.

7.3 Policy Interpretation

- Cultivated area and production are not identical measures; a state with a large cultivated area does not automatically have the largest production in the displayed comparison.
- The long-term rise in yield suggests that productivity is an important part of agricultural performance, alongside the amount of land cultivated.
- Regional differences mean that agricultural policies may need to account for state-specific crop patterns rather than using one approach for every region.

8. Key Findings

Finding	What the dashboard shows
1. Yield shows a long-term upward pattern.	The yield line moves from about 3.8 in 1997 to around 5.0 near 2019, with several fluctuations in between.
2. Production has grown compared with the late 1990s.	The year-wise bars in the later part of the series are generally much higher than the earliest years.
3. Rice and wheat dominate cultivated area.	The crop chart shows a clear gap between rice/wheat and the other crops included in the comparison.
4. State-level differences are substantial.	The state charts show large differences in both cultivated area and agricultural production.
5. The dataset contains mixed production units.	The unit-type chart is dominated by Nuts, with smaller shares for Tonnes and Bales, so unit definitions matter.
6. The dashboard supports different kinds of questions.	The same data can be read differently by a farmer, an investor or a policy analyst.

9. Limitations

- The report uses the supplied dashboard screenshots rather than the underlying raw dataset.
- Some charts use aggregated values and dashboard-specific units, so they should not be treated as raw farm-level measurements.
- The report describes patterns visible in the dashboard; it does not establish causes for changes in production or yield.
- The 'Whole Year' category dominates the season chart, so seasonal comparisons need to be made carefully.
- Investment-related interpretation is limited because the dashboard does not contain prices, costs, returns or financial-market data.

10. Conclusion

The India Agricultural Performance Dashboard provides a useful visual summary of agricultural production, cultivated area and yield across years and states. The main pattern is a generally improving yield over the long period, along with higher production levels in the later years. At the same time, the state and crop charts show that agricultural activity is not evenly distributed.

The three dashboard viewpoints make the project more practical. A farmer can focus on crops, seasons and yield; an investor can focus on production scale and regional concentration; and a policy analyst can focus on state-level gaps and long-term trends. Overall, the dashboard turns a large agricultural dataset into information that can be understood quickly and compared visually.